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3. Flow control in Java

3.1 Decision making statement :

Decision-making statement allows you to execute different code blocks based on certain conditions. In Java, you have two main decision making statements:

if statement
if-else statement
else-if statement (Ladder)

if statement

The 'if' statement is used to execute a block of code if a given condition is true. It has the following syntax:

```
if (condition) {
    //code to execute if the condition is true
}
```

Example:

```
int number = 10;
if (number > 0) {
    System.out.println(" Number is positive");
}
```

if-else statement :

The 'if-else' statement extends the 'if' statement by allowing you to execute different blocks of code based on

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whether a conditions is true or false.

```
if (condition) {
    // code to execute if the condition is true
} else {
    // code to execute if the condition is false
}
```

Example:

```
int num = -5;
if (num > 0) {
  System.out.println("Number is positive");
} else {
  System.out.println(" Number is non-positive");
}
```

else-if Ladder:

You can also use an 'else-if' ladder to test multiple conditions sequentially.

```
if (condition1) {
    // code for condition1
} else if (condition2) {
    // code for condition2
} else if (condition3) {
    // code for condition3
} else {
    // code if none of the conditions are true
}
```

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Example :

```
int score = 85;
if (score >= 90){
  System.out.println("Grade: A");
}else if (score >= 80){
  System.out.println("Grade: B");
}else if (score >= 70){
  System.out.println("Grade: C");
}else{
  System.out.println("Grade: F");
}
```

3.2 Switch statement:

The 'switch' statement in Java is used to evaluate an expression against a series of constant values. It provides an efficient way to perform different actions based on the value of the expression. Here's the syntax of the 'switch' statement:

```
switch(expression){
  case value1:
      //code to execute if expression equals value1
    break;
  case value2:
     // code to execute if expression equals value2
    break;
  case valueN:
    // code to execute if expression equals valueN
    break;
  default: // code to execute if none of the cases match expr.
  }
```

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Example:

```
int day = 3;
string dayName;

switch (day) {
  case 1:
    dayName = "Monday";
    break;
  Case 2:
    dayName = " Tuesday";
    break;
  Case 3:
    dayName = " Wednesday";
    breack;
  default:
    dayName = "Invalid day";
}
System.out.println ("Day :" +dayName);
```

3.3 Loops in Java :

Loops allow you to repeatedlly execute a block of a java.code. Java supports three types of loops:

1) For loop

2) While loop

3) do-while loop

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For loop:

The 'for' loop is used for iterating a specific numbers of times.

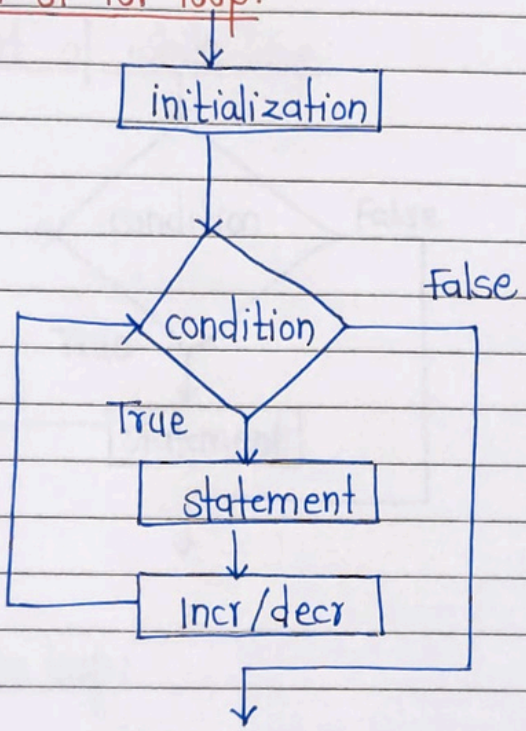
syntax:

```
for(initialization; condition; update){
   //code to execute in each iteration
}
```

Example:

```
for (int i=1; i<=5; i++){
   System.out.println ("Iteration" +i);
}
```

Flowchart of for loop:



While loop:

The 'While' loop repeatedly executes a block of code as long as a given condition is true.

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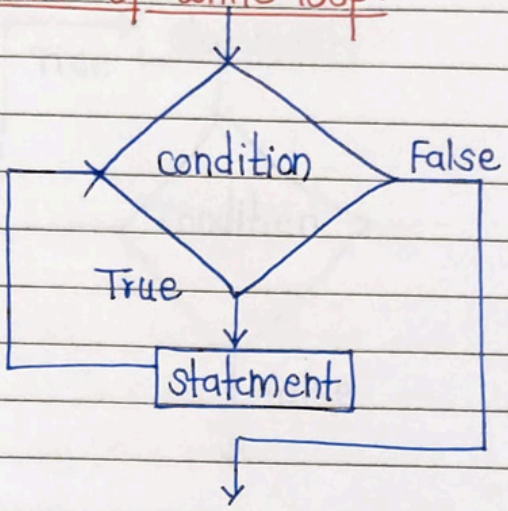
syntax:

```
While(condition){
 // code to execute as long as the condition is true.
}
```

Example :

```
int count = 0;
while(count<5){
  system.out.println("count: "+count);
  count++;
}
```

Flowchart of while loop:



do-while loop:

The 'do-while' loop is similar to the 'while' loop, but it ensures that the code block is executed at least once before checking the condition.

syntax:

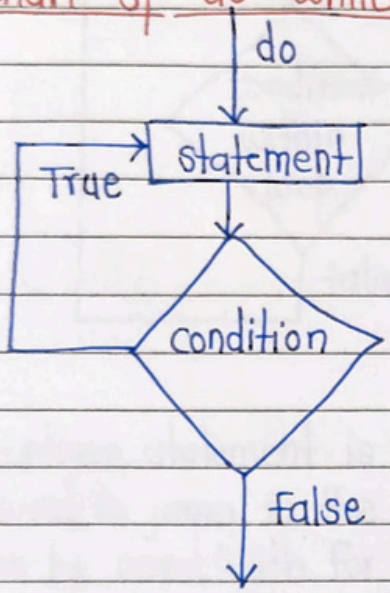
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```
do{
 // code to execute at least once
}while(condition);
```

Example:

```
int x=1;
do{
  system.out.println("value of x: "+x);
  x++;
}while(x<=5);
```

Flowchart of do-while loop:



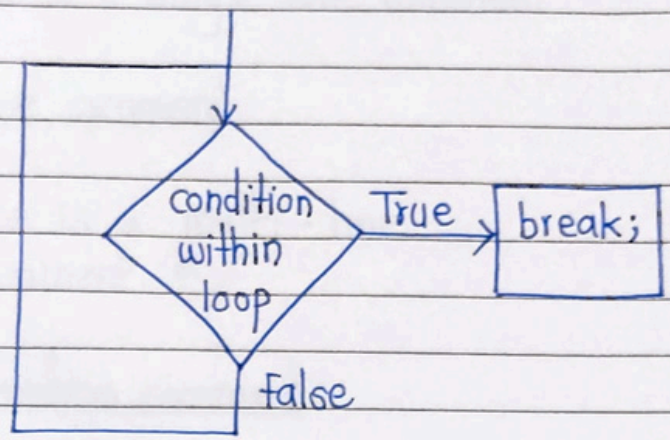
3.4 Break and continue statement

When a break statement is encountered inside a loop, the loop is immediately terminated and the program control resumes at the next statement following the loop. The Java break statement is used to break loop or switch statement. It breaks the current flow of the program at specified condition. In case of inner loop, it breaks only inner loop.

Example:

```
for (int i = 1; i<=10; i++) {
  if (i == 6) {
    break; //Exit the loop when i equals 6
  }
  system.out.println("value of i: " +i);
}
```

Flowchart of Break statement



The continue statement is used to loop control structure when you need to jump to the next iteration of the loop immediatly It can be used with for loop or while loop. The Java continue statement is used to continue the loop. It continues the current flow of the program and skips the remaining code at the specified condition.

Example

```
for (int i =1; i<=10; i++) {
  if (i== 5) {
    continue; // It will skip the rest statement
  }
  System.out.println(i);
}
```

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8.5 Comments in Java :

Comments are non-executable statements that provide explainations within the code. They are used to improve code readability and understanding. Java supports three types of comments.

Single-Line Comment :

```
// This is a single line comment
```

Multi-line comment :

```
/* This is a multi-line
   comment */
```

Documentation comment :

```
/**
 * This is a documentation comment.
 * It is used for generating documentation.
*/
```